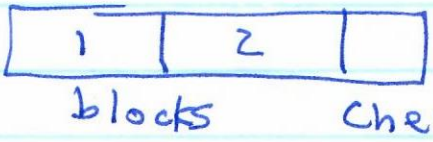
# Ch 11 serial Communication

Serial communication - information transmitted from a source to a destination over a single pathway (1 bit at a time)

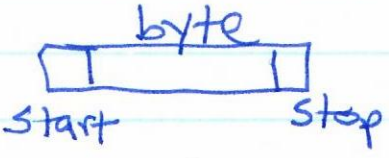
2 types:

1) synchronous transmission
   - transmit a block of data and a clock signal or a known bit pattern that is used for synchronization

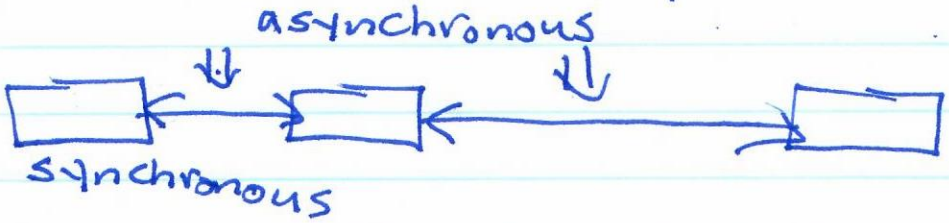



   - blocks ⇒ serial stream of bits
   synchronization is needed between sender & receiver

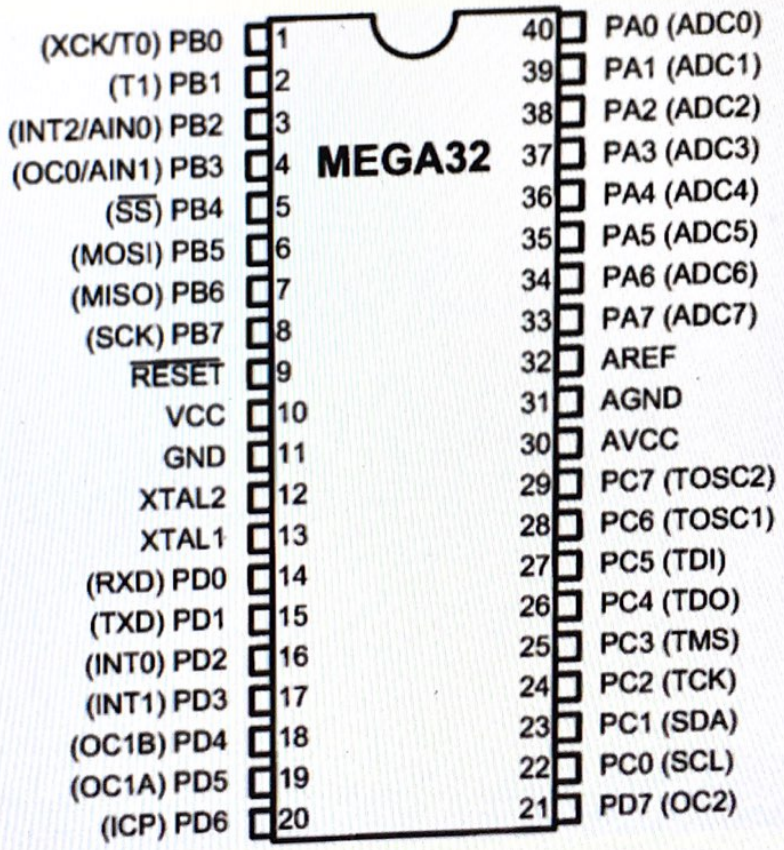
2) asynchronous transmission
   - transmit 1 character at a time by adding start & stop bits to the character code



asynchronous at the byte level



# Chapter 4-Fig4-1



Serial transmission modes - defining the direction of signal flow between 2 linked devices

3 modes:

1) simplex transmission - data flows in 1 direction only
   - radio
   - GPS

2) half-duplex transmission - data can flow in both directions just 1 at a time
   - 1 wire
   - walkie talkie
   - printer

3) full-duplex transmission allows data to flow in both directions concurrently (2 wire)
   - phone
   - Internet

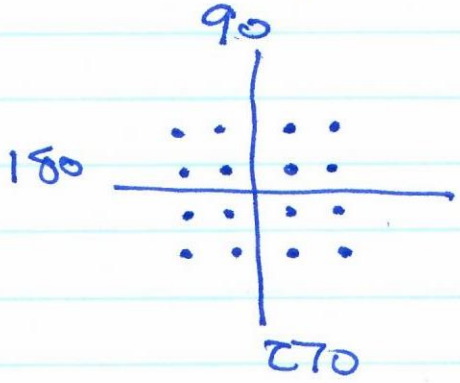
baud rate - symbols per second

1 symbol = 1 bit ⇒ bits per second

Ex) cable signal

QAM (quadrature amplitude modulation)

QAM-16




phase, magnitude

2^4 (3 data bits + 1 error bit)

---

Overhead

COM PORT { 8 bits (data), start bit, stop bit, parity bit

IBM & compatibles

% overhead

$$\frac{\text{start, stop, parity}}{8+1+1+1} = \frac{3}{11} \approx 27\%$$

$$\frac{\text{start, stop}}{8+1+1} = \frac{2}{10} = 20\%$$

USART (Universal Synchronous/Asynchronous Receiver/Transmitter in the AVR

4 features:
- normal asynchronous
- double-speed asynchronous
- master synchronous
- slave synchronous

Synchronous
- ADC
- EEPROM communication

asynchronous
- COM PORT x86
- full-duplex capability

AVR registers associated with USART
- UDR (USART Data Register)
- control registers: UCSRA, UCSRB, UCSRC
- UBRR (USART Baud Rate Register)

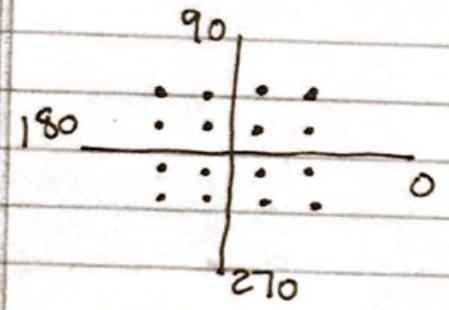
band rate - Symbols per second

bits per second (bps)

Ex) cable signal

QAM (quadrature amplitude modulation)

QAM-16




phase, magnitude

$2^4$ (3 data bits + 1 error bit)

---

overhead

COM Port IBM & Compatibles { 8 bits, Start bit, Stop bit, parity bit

9600

% overhead

Start, stop, parity

$\frac{1+1+1}{8+1+1+1} = \frac{3}{11} \Rightarrow 27\%$

Start, Stop

$\frac{1+1}{8+1+1} = \frac{2}{10} \Rightarrow 20\%$

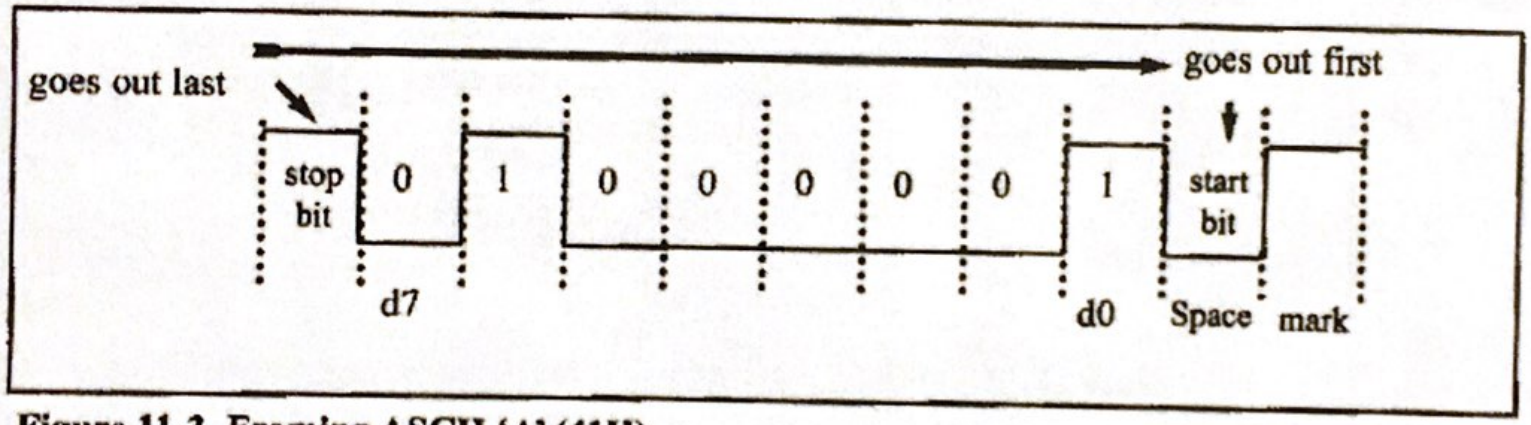
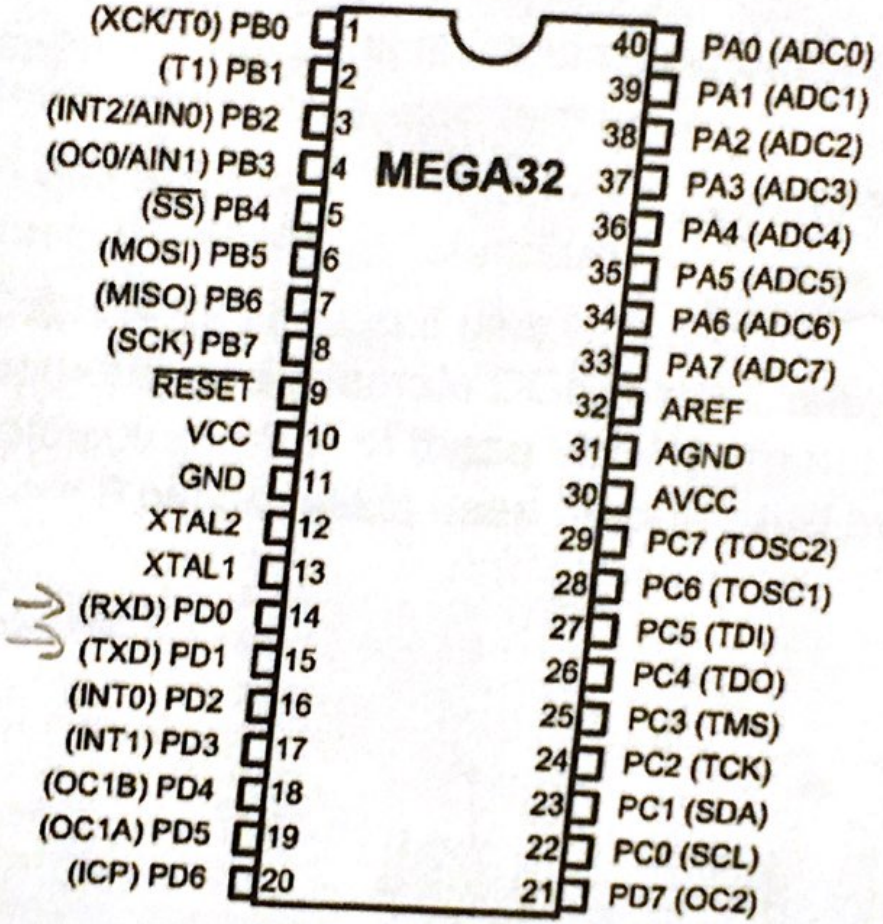


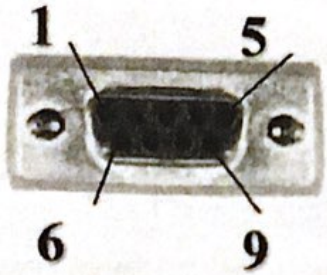

**Figure 11-3. Framing ASCII 'A' (41H)**

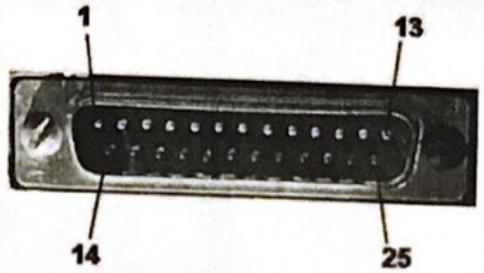
# Chapter 4-Fig4-1



# RS-232C

- TTL logic signals are not what is transmitted between equipment such as modems, computers, and terminals.
- One standard that specifies the signals for asynchronous communication is the RS-232C standard.
- The RS-232C specifies 25 signal pins with a male DTE (Data Terminal Equipment) and a female DCE (Data Communications Equipment) connector. Most of these signals are not used in most applications, so a 9 pin subset is used (popularized by the IBM PC or compatible).



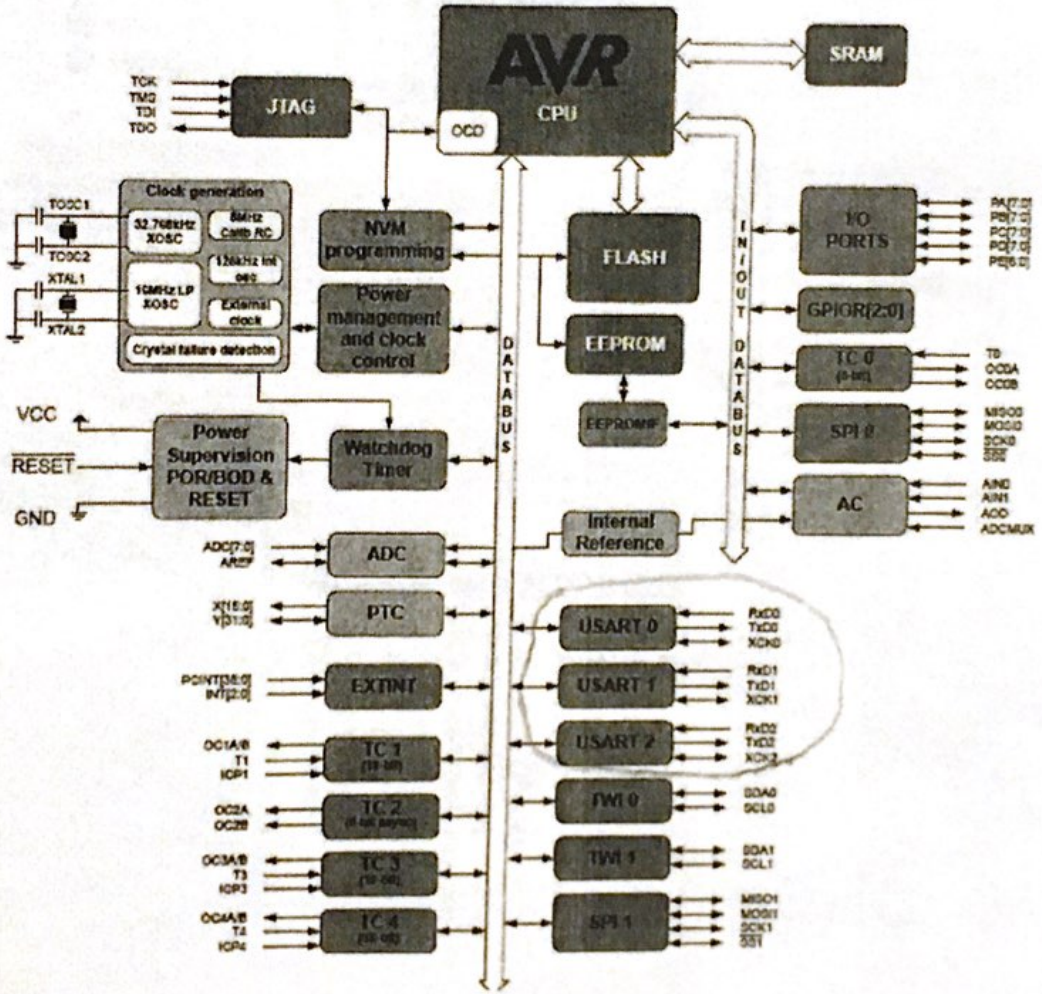
# Important Asynchronous Signals in RS-232C

| Description | PIN # (9-PIN) | From | Abbreviation |
|---|---|---|---|
| **Data Leads** | | | |
| Transmit Data | 3 | DTE | TD |
| Receive Data | 2 | DCE | RD |
| **Power On Indicator Leads** | | | |
| Data set ready | 6 | DCE | DSR |
| Data terminal ready | 4 | DTE | DTR |
| **Leads that announce that an outside event has taken place** | | | |
| Data carrier detect | 1 | DCE | CD |
| Ring Indicator | 9 | DCE | RI |
| **Ready to send/receive handshake leads** | | | |
| Request to send | 7 | DTE | RTS |
| Clear to send | 8 | DCE | CTS |
| **Ground leads** | | | |
| Signal ground | 5 | | SG |

# ATmega324pb

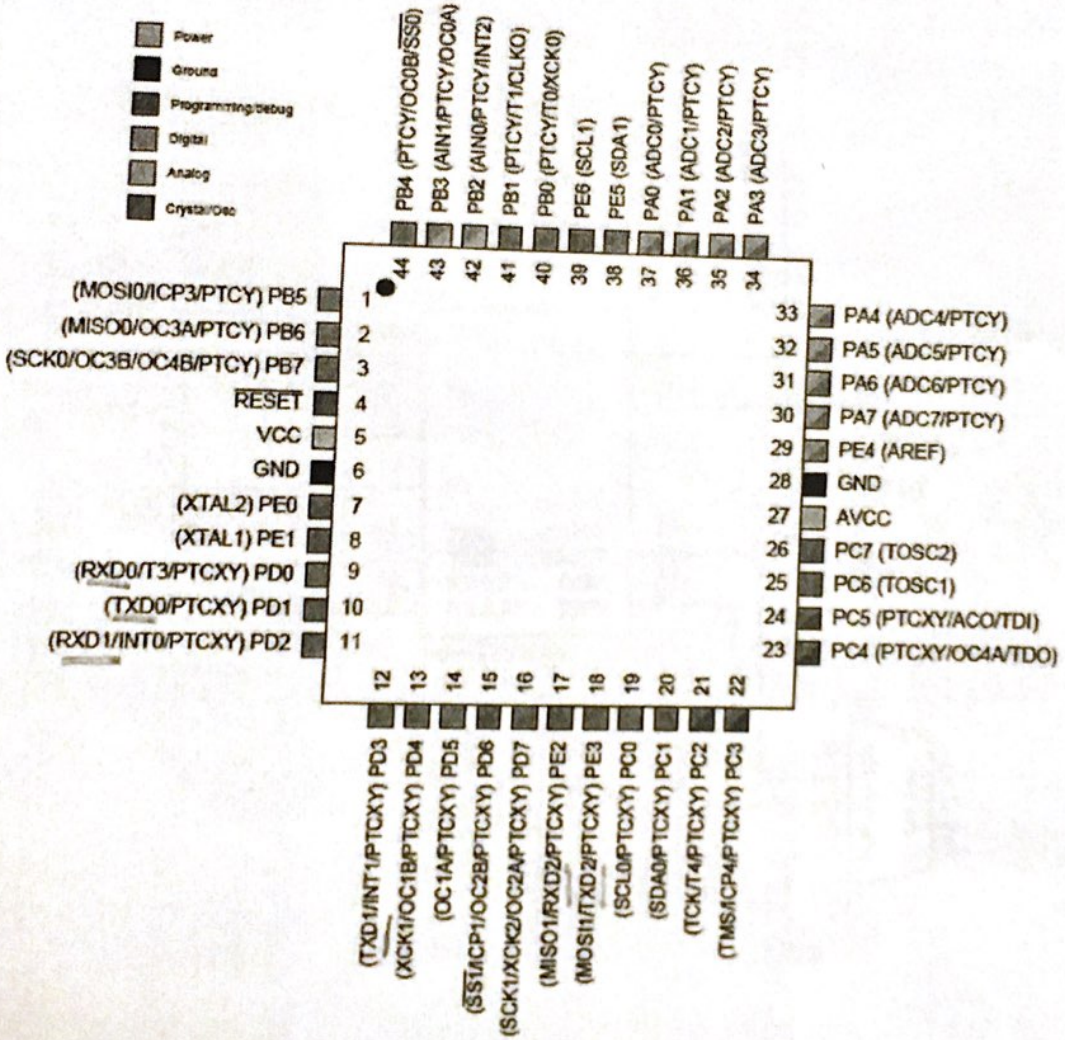
## Block Diagram

**Figure 4-1. Block Diagram**

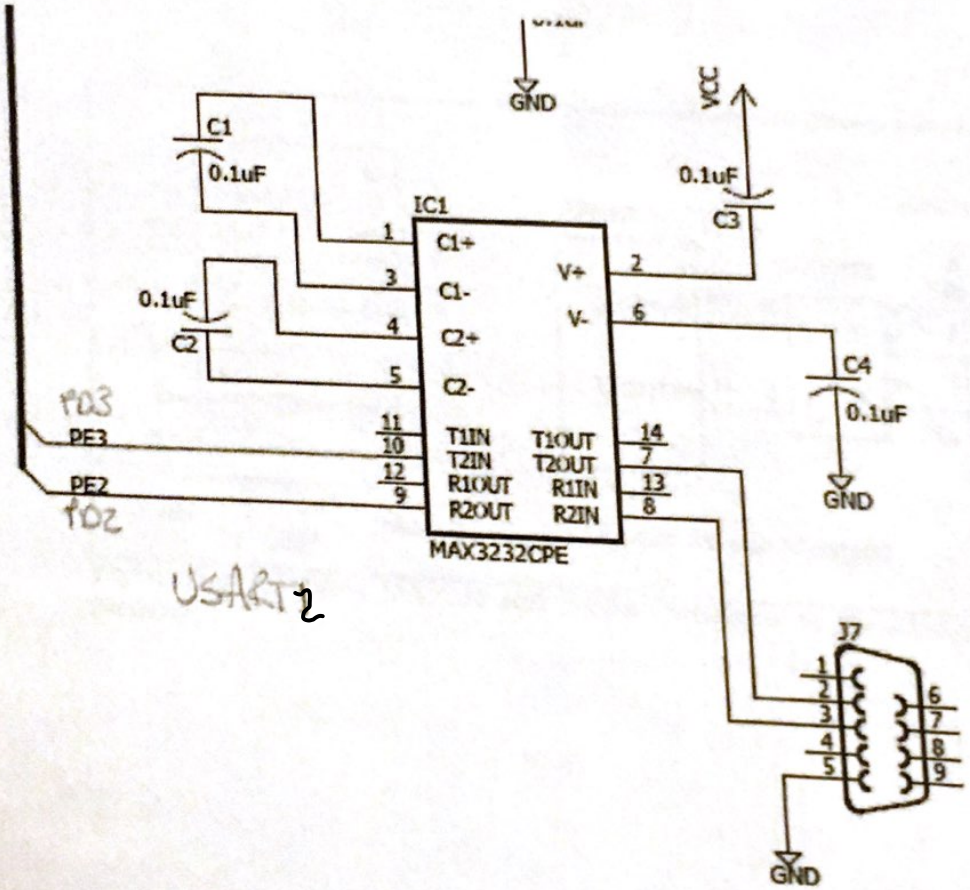


# ATmega324pb

**Figure 5-1. Pinout ATmega324PB**



# ATmega324pb



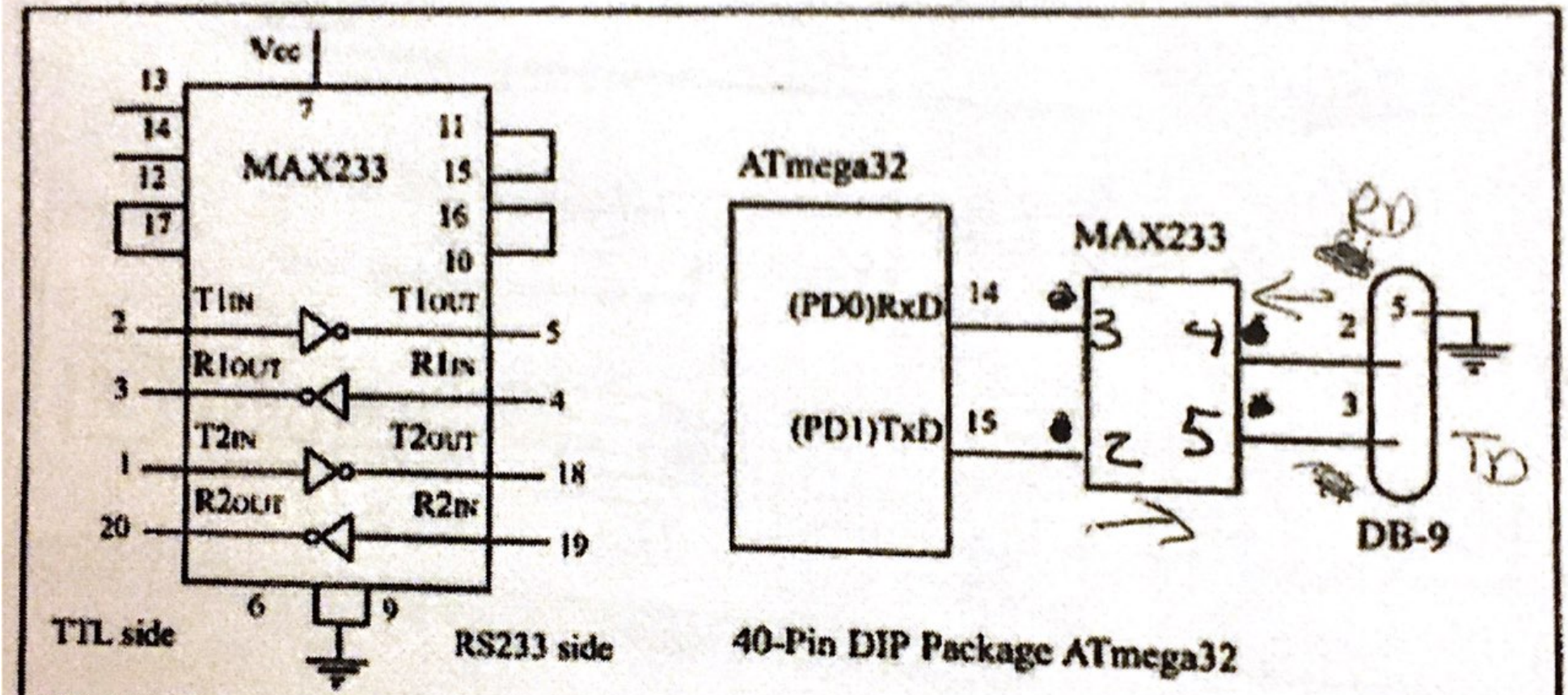



**Figure 11-8. (a) Inside MAX233 and (b) Its Connection to the ATmega32 (Null Modem)**